

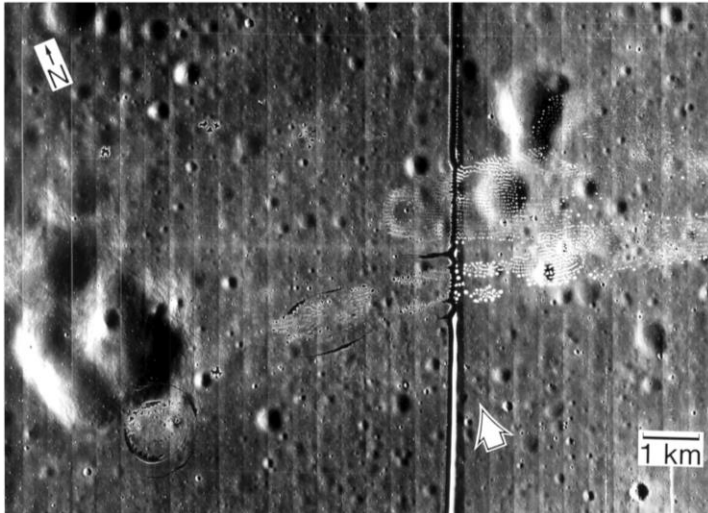
Original Painting



Filtered Painting



Original Image (Apollo 11)



Stitched Image



Properties and Applications of the 2D Fourier Transform (II of II)

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Objectives

- To determine the behavior of sinusoids with varying parameters in Fourier space and their superposition.
- To investigate the properties of convolution involving Dirac-delta functions in two dimensional space
- To isolate and remove repeating sinusoids in an image such as in a canvas painting, and in stitched photographs by manipulations in the Fourier space.

Rotation Property of the Fourier Transform

The Fourier Transform (FT) of a function gives its spatial frequency distribution – it acts like a histogram of the sinusoid frequencies that make up the function. If we are given with a 2D function, its rotation causes the Fourier Transform to also rotate in its own space.

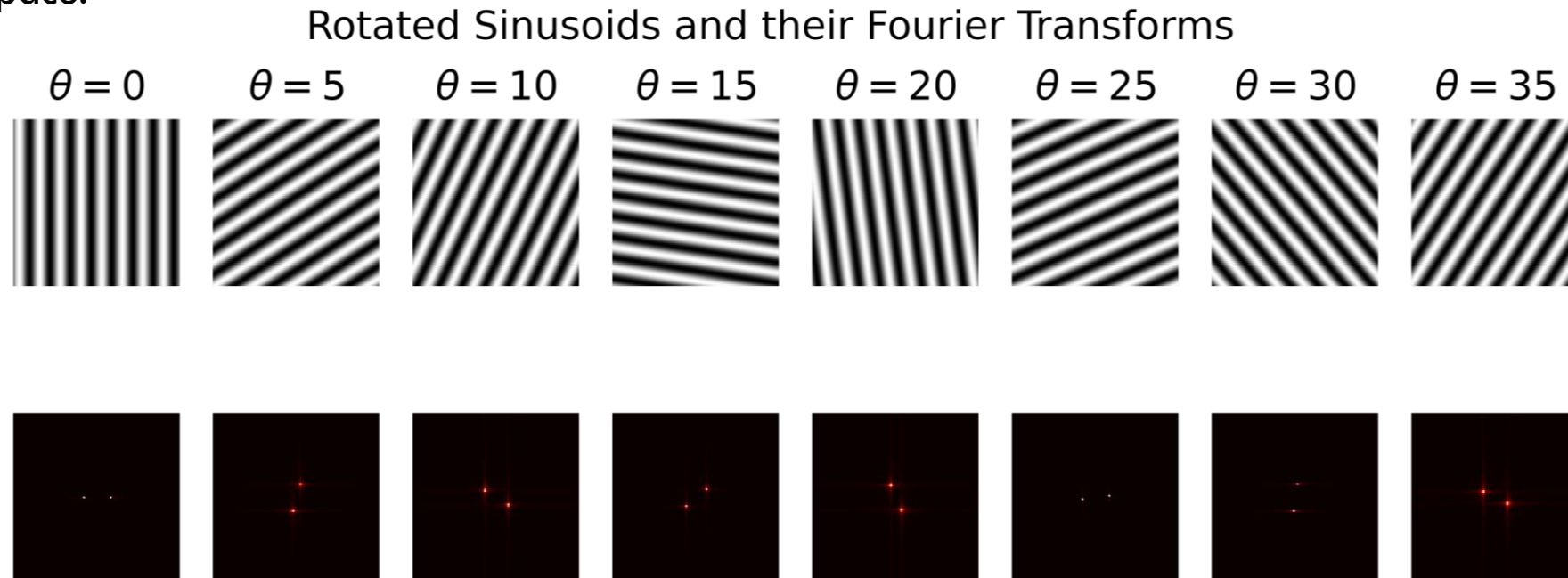


Fig 1. A sinusoid of constant frequency rotated along the center and its FT

We can see from Fig 1. that the Fourier transform of a sinusoid manifests as two dots which are symmetrical about the centroid of the image. Upon rotation of the sinusoid in real space, an accompanying rotation in the Fourier space follows. What would happen if we superimpose two sinusoids together and varied their frequencies?

Rotation Property of the Fourier Transform

Constant Sine along Y Multiplied to Rotating Sine along X

$\phi = 0$ $\phi = 5$ $\phi = 10$ $\phi = 15$ $\phi = 20$ $\phi = 25$ $\phi = 30$ $\phi = 35$

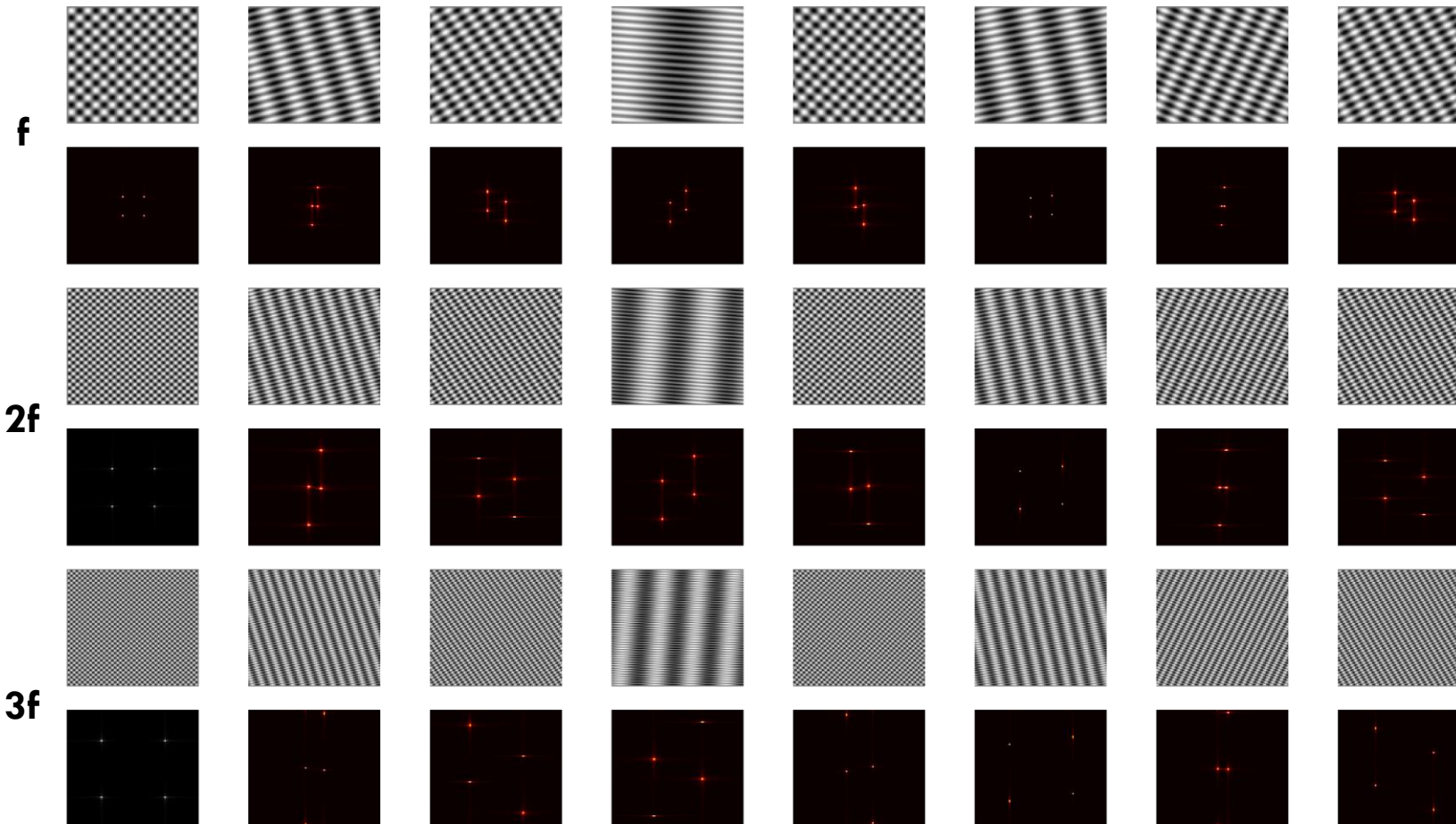


Fig 2. A sinusoid superposition with varying frequencies.

We can see that the appearance of two sinusoids results in two pairs of dots in Fourier space. If we add more sinusoids, then there will be an equal number of dot pairs added in Fourier space. Each sinusoid and their respective FTs rotate independent of each other despite being in a superposition – this is a very useful attribute.

If we want to remove sinusoidal artifacts in an image, we look for pairs of dots in Fourier space and remove them! We can use this fact to remove weaves, since as we can see from Fig 2., these types of textures are only due to sinusoids superimposed together. Any repeating pattern in an image that act sinusoidal in nature can be removed by filtering them out of Fourier space with dots.

Canvas Weave Modeling and Removal

Since we now know that any repeating pattern in an image is just a combination of different sinusoids (which makes sense if you'd taken a Math course as sinusoids are a complete set), then we can do manipulations in Fourier space to remove or enhance them. In this demonstration, I remove the texture of the canvas in the painting in Fig 4. in order to investigate its brush strokes. Now, I do not have any skill whatsoever when it comes to painting - but these kinds of algorithms can help more skilled visual artists in their various crafts, even in detecting whether a painting is an original or a replica.

Original Painting

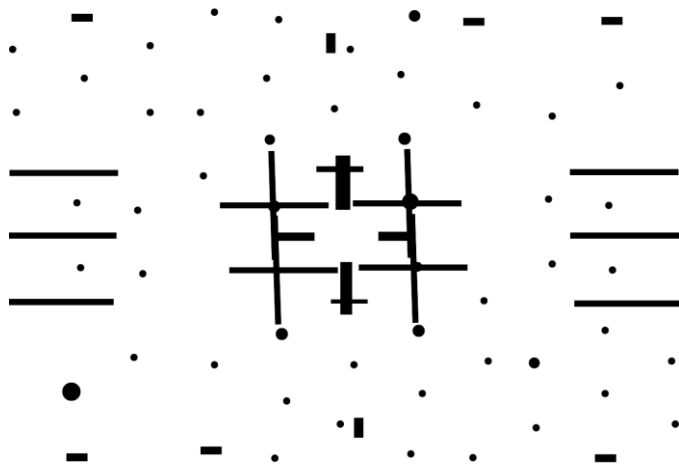
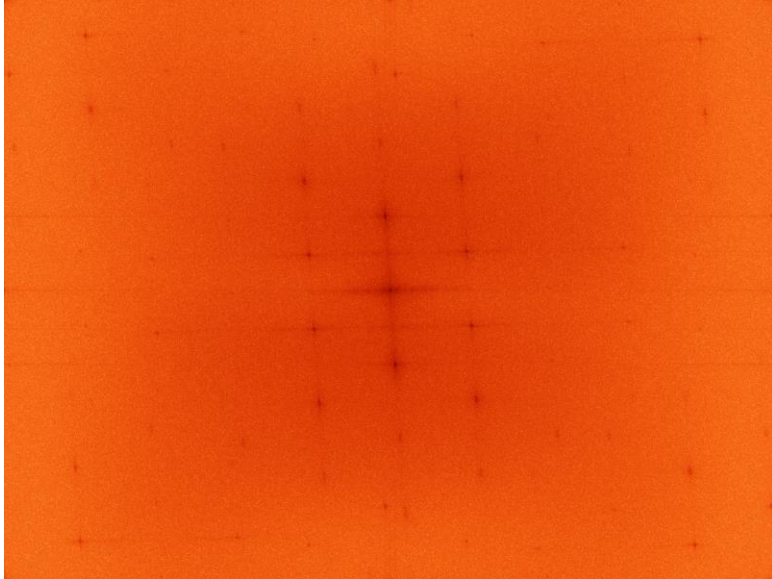


Filtered Painting



Fig 4. Canvas Pattern Removal in a Painting (Artwork by Dr. Vincent Daria)

Canvas Weave Modeling and Removal



In Fig 5., I display the Fourier transform of Dr. Daria's painting in a logarithmic scale in order to see faint dots near the periphery of the space. As expected, the Fourier space is littered with dot pairs due to the sinusoid superposition that comprise the canvas weaves. On the other hand, the image data's Fourier transform in this figure is the dark spot at the middle. This contains all information regarding brushstrokes, colors, and all other features we want to preserve.

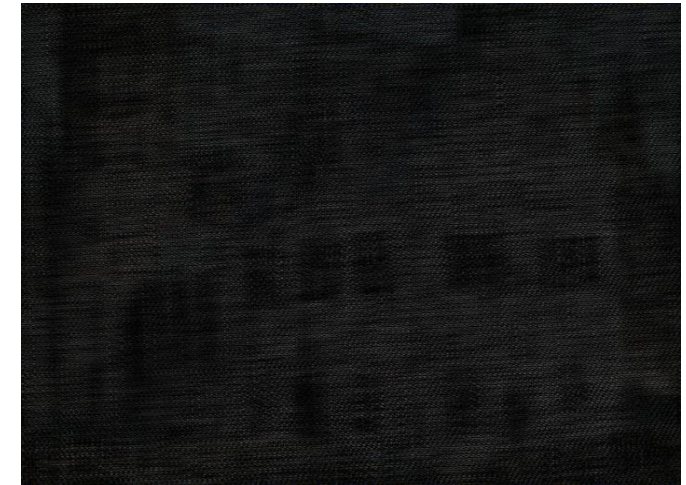


Fig 6. Canvas Pattern of Dr. Daria's Painting

In order to remove the canvas pattern in the painting, we convolve the two images in Fig 5. However, since both images are already in the Fourier space this mathematical operation becomes much simpler – we simply multiply the two functions in Fourier space! To recover the resultant image, we just take the inverse FT of the resulting product. Note that if we reverse the colors in the mask used (Fig 5, bottom) we will instead highlight the canvas weaves and remove the painting itself as seen in Fig 6.

Fig 5. FT of Dr. Daria's Painting in a Logarithmic Scale (top) and Mask used to Remove Canvas Pattern (bottom)

Convolution Theorem Redux

In the previous demonstration, we used the convolution operator in Fourier space in order to apply the created mask to the image's Fourier transform through simple elementwise multiplication. Here, we apply the same operator in functions existing in real space (the images themselves) to study its properties.

From the first demonstration (rotation of sinusoids), we can infer that the FT of two dots is a sinusoid as seen in (Fig 7, top) since the FT has an inverse operator. The white pixel in the middle is due to an unavoidable division by zero, which we disregard. We also know from the second activity this semester that the FT of a single circle is a circular diffraction pattern.

Curiously, when we put circles on top the supposed location of the dots (**Fig 7, bottom**) the FT looks like the product of a sinusoid and a circular diffraction pattern. Why is that?

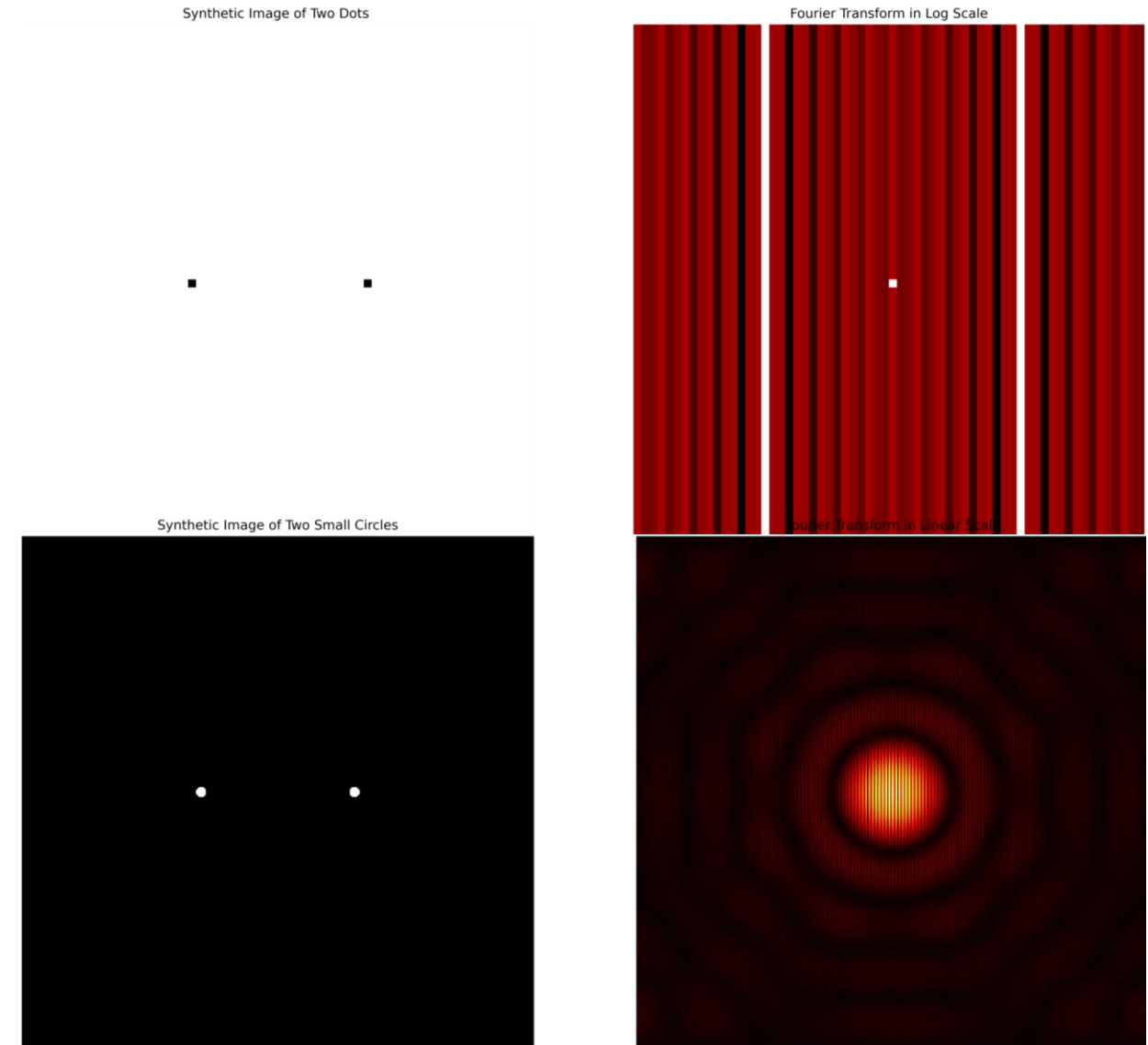


Fig 7. Fourier transform of two pixels (top) and two small circles (bottom)

Convolution Theorem Redux

The convolution of two functions in the real space is just the product of their FTs i.e.

$$f ** g = FG.$$

An interesting property is that if we convolve a Dirac-delta and some function in 2D space, the result is just the replication of the function the location of the Dirac-delta

$$\int_{-\infty}^{+\infty} \delta(x - x_o - x', y - y_o - y') f(x', y') dx' dy' = f(x - x_o, y - y_o)$$

The “proof” for this is the previous slide. The function we convolve to the Dirac-delta is a single circle. When we multiply their FTs (a sinusoid and a circular pattern, respectively), we get the pattern in (Fig 7, bottom). Taking the inverse FT will yield as two circles, one for each location of the Dirac-delta. Even in a computational physics class we can do proofs, although not as rigorous! This can extend to any number of Dirac-deltas, as seen in Fig 8.

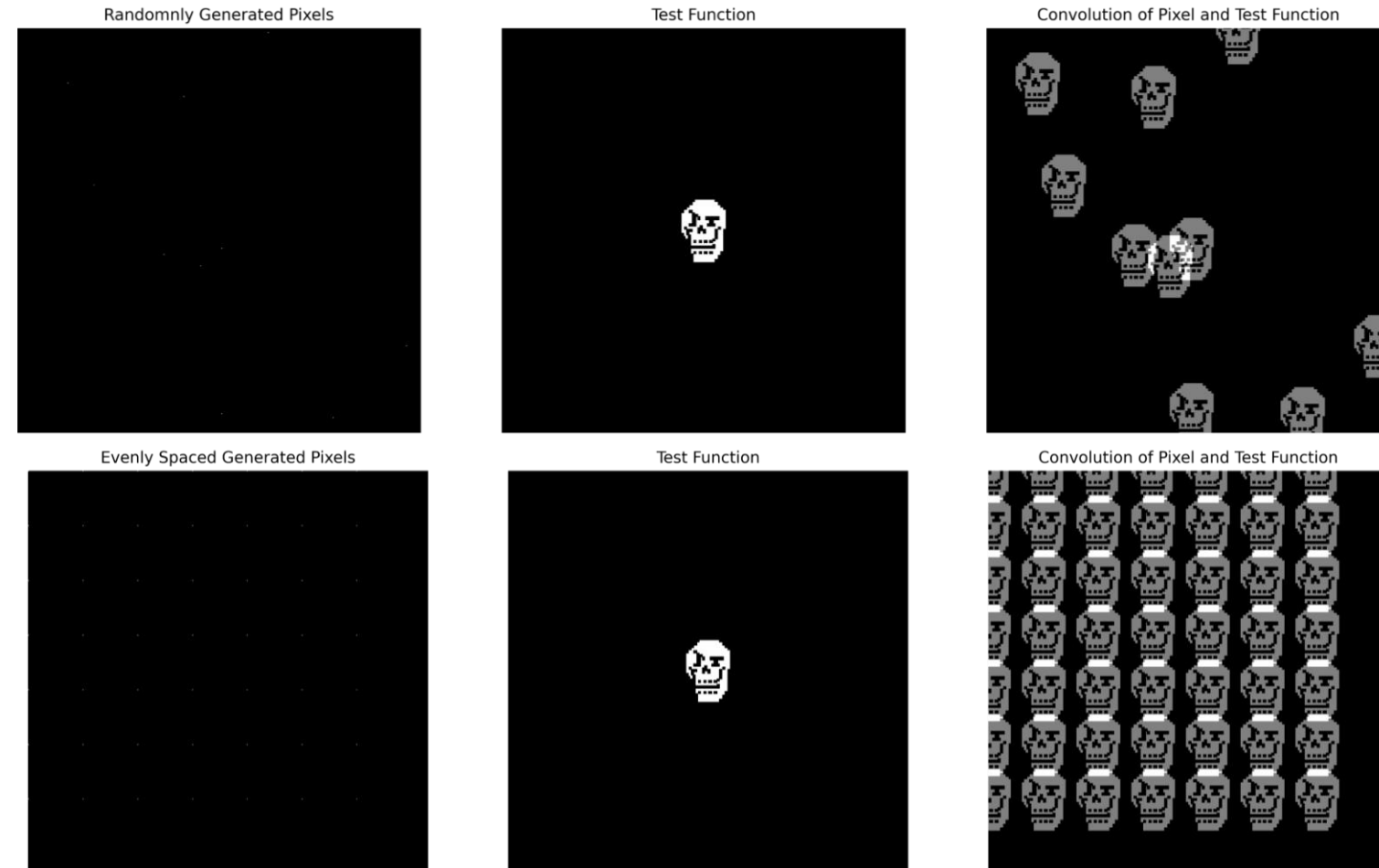


Fig 8. Convolution of randomly (top) and evenly (bottom) spaced Dirac-deltas with a 2D function

Fingerprint Ridge Enhancement

From the previous activities, we have explored various properties of sinusoids and their FTs – along with applications to canvas removal. Here, we use the same techniques to highlight the ridges of a fingerprint. To be clear, the ridges refer to the lines that make up the fingerprint... The fingerprint itself is sinusoidal in nature with frequencies that are not varying by much, as we can see from the image. The ridges are reasonably spaced apart evenly while wrapping around the middle of the finger. Hence, its manifestation in Fourier space is a collection of “dots” rotated along angles in roughly the same region away from the center (since frequency is roughly the same). We can see this as a “circular disk” pattern in the third image in Fig 9.

Instead of suppressing these regions in Fourier space, we amplify them cause we want to isolate ridges, not suppress them. Hence, we use the filter as shown in Fig 9 to “delete” everything except the ridges. Note that as always, the center contains image data and must more or less always be preserved. **In fact, we can apply the procedure again to the enhanced fingerprint image to binarize it, further revealing the ridges for use in softwares!! As an added bonus, we can just apply the same mask!**

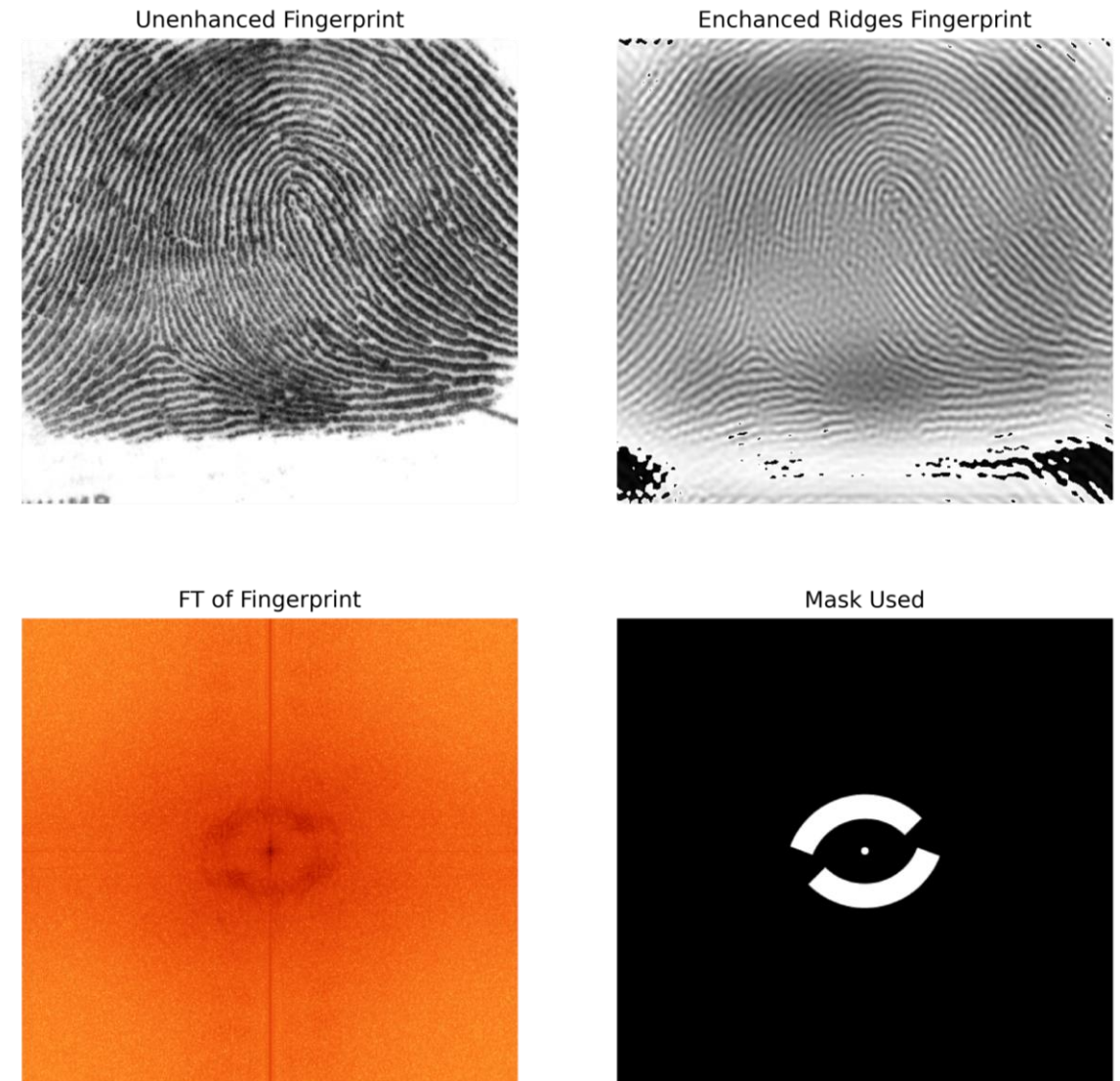


Fig 9. Fingerprint Ridge Enhancement (not yet binarized)

Lunar Landing Scanned Pictures: Line Removal

As a culmination of everything in this activity, we process images from the original Apollo 11 moon landings. Due to the limited technology of the time, imaging systems in the lunar module were not able to capture a large area. The camera then took several images which were then stitched together back here on Earth. However, this method introduced repeating stitch marks on the images which are undesired artifacts. **The challenge is now this: how do we remove these stitch marks without losing or adding relevant image data?** The answer is of course 1) convolve the FT and the fft-shifted mask in Fourier space 2) profit. In this case, the FT of the Apollo image is a “crosshair”. We can confirm this by taking the FT of the mask, it will return the stitch marks.

This activity is a testament to the reason why we study the Fourier domain in physics so much. It's a very convenient vector space to work in and it simplifies so many problems (in this case, the convolution operator). Truly one of the methods of all time (frfr)!!!

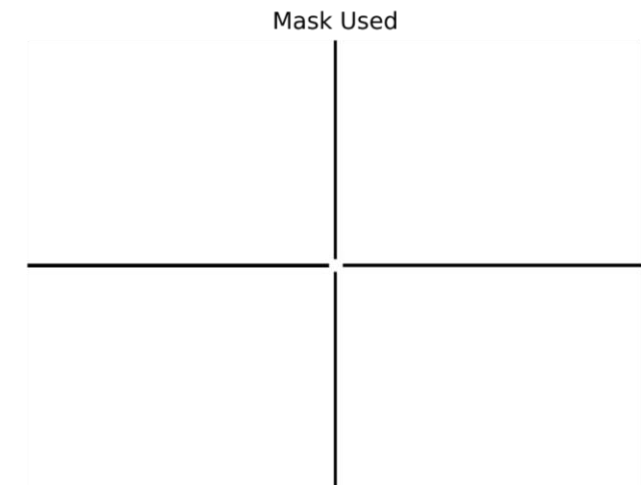
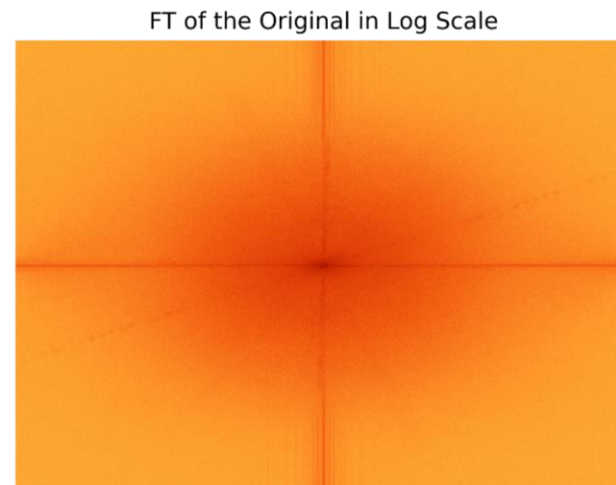
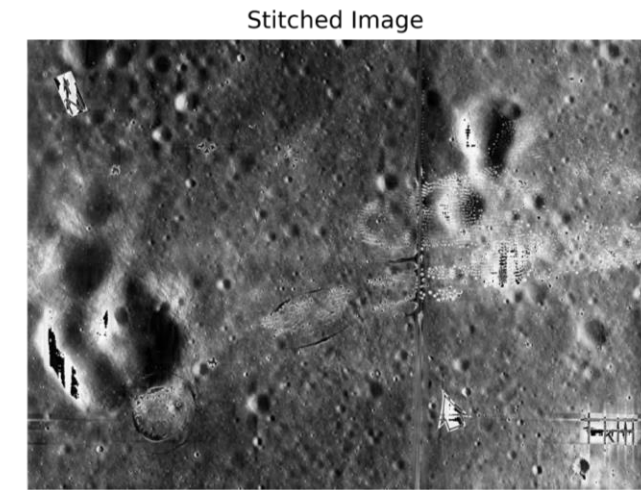
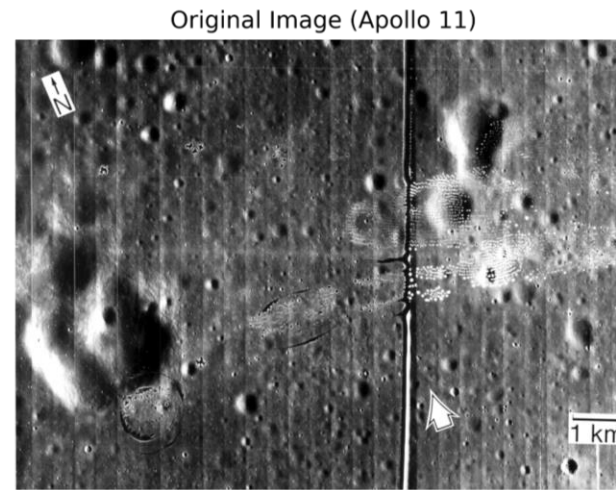


Fig 9. Enhancing images from the moon landing by removing stitch lines

Key Takeaways

- A sinusoid will yield two symmetrical dots in the Fourier space with distance dependent on frequency and angle depending on the sinusoid's rotation. If we superimpose sinusoids, we can basically model canvas weaves and other repeating patterns in images.
- To remove unwanted sinusoids in an image, delete them in the FT space by convolving with mask where they are marked 0. To preserve the sinusoids, convolve with a mask where they are marked as 1's. However, the quality of the filtered image still heavily relies on the quality of the mask itself
- The convolution of a Dirac-delta with any other function replicates that function in the position of the delta.

Reflection

This activity finished the module on Fourier transforms, and I can say that it gave me a first hand understanding on what the Image and Video Processing Laboratory does for their research. Of course I always knew that they dealt with tracking objects and image data in general, but I feel like this module made me appreciate their work in a deeper (yet still introductory) way. The techniques used in this activity were actually mathematically simple – this would only qualify as a peripheral problem at best on a Mathematical Methods class. However, actually implementing it beyond theory and encountering bugs, undesired results (I use more than 10 wrong masks before I arrive at the correct one, see my github repo!), and other challenges made this endeavor very satisfying to accomplish.

I was able to provide all expected outputs, and even went beyond on some parts such as in Fingerprint Isolation. **I'm not saying that my work is perfect – but due to the effort I exerted and my results I think I deserve a 100/100 at last!**

References

- [1] Soriano, M. (2023). ACTIVITY 2 (PART 2 OF 2) – PROPERTIES AND APPLICATIONS OF THE 2D FOURIER TRANSFORM. National Institute of Physics, UP- Diliman.
- [2] Sprites for Papyrus of the videogame Undertale in <https://www.canva.com/> education edition.